SVKM'S NMIMS Deemed to be UNIVERSITY | SHIRPUR
MUKESH PATEL TECHNOLOGY PARK

Pranav Mundhra

B.Tech Artificial Intelligence & Machine Learning | 2027 Batch



SUMMARY

Final-year AI/ML student who turns ideas into working, tested software. Projects include an LLM-based recommendation chatbot, a federated-learning capstone on wind-turbine data, and a system that shares ML predictions across two computers. Built a Hindi/English/Marathi translation pipeline as a Backend & AI Intern at Finkeep, and led the team that won AISEHack 2026.

EDUCATION

SVKM's NMIMS MPSTME, Shirpur **Jul 2023 – 2027**
B.Tech, Artificial Intelligence & Machine Learning (CGPA: 3.41 / 4.00)

Centre Point School, Dabha, Nagpur **Apr 2021 – Apr 2023**
CBSE Class XII (PCM): 76.60% | CBSE Class X: 86.40%

SKILLS

- **Languages & Databases:** Python, SQL, PostgreSQL (PL/pgSQL procedures, functions, triggers), Neo4j / Cypher
- **GenAI & NLP:** LLMs (Gemini, Mistral), prompt engineering, chatbots, RAG, LangChain, Pydantic, sentence embeddings, IndicTrans2
- **ML & Data:** PyTorch, TensorFlow, Keras, Scikit-learn, Flower (federated learning), NumPy, Pandas, ONNX
- **Engineering & Tools:** FastAPI, pytest, GitHub Actions, Git, Docker, Linux, Streamlit, Supabase
- **Leadership:** Treasurer, ACM Student Chapter (since Nov 2025); team lead, AISEHack 2026 winning team

WORK EXPERIENCE

Finkeep — Backend & AI Intern
Client Project: ICAI Nagpur Branch – Institutional web platform **May 2026 – June 2026**

- Built the platform's translation pipeline (my main contribution), which converts content between Hindi, English and Marathi using IndicTrans2.
- Designed the initial PostgreSQL database (Supabase), with role-based access for 8 user types and 40+ features, for CA members, students, employers and branch staff.

MAJOR PROJECTS

Emotion-Based Anime Recommender — LLM Chatbot **Oct 2025**

- Built a chatbot (LangChain, Gemini/Mistral) that asks questions over 6 turns to learn the user's mood and preferred genres and themes.
- Used Pydantic to make the LLM return valid JSON, with safe fallbacks; ranked 50 anime from the Jikan API by sentence-embedding similarity and had the LLM explain the top picks.

Federated Fault Detection for Wind Turbines (Team Capstone, paper in prep) **Jun 2026 – Present**

- Trained anomaly-detection models (autoencoders with Flower and PyTorch) across 22 wind turbines without sharing their raw data; the federated model matched per-turbine training on 58 real events (CARE score 0.634 vs 0.627).
- Made every result reproducible with a data-leakage check, checksum-verified datasets, five-seed runs with confidence intervals, a command-line tool and 70+ automated tests in CI; code archived on Zenodo.

Distributed ML Inference Platform **Oct 2026**

- Deployed an ML prediction system across two PCs on a LAN: a FastAPI gateway checks each request and sends it to the next available worker (round robin); each worker holds full copies of 3 models.
- Added automatic worker registration, heartbeat-based failure detection (a silent worker is dropped after 8 s and rejoins on restart), request tracing with Lamport clocks and a live dashboard; covered by 361 tests.

OTTools — Optimisation Toolkit (PyPI) **Apr 2025**

- **Co-authored OTTools on PyPI**, a package for solving linear-programming and transportation problems (Simplex, Big-M, VAM, MODI), and built its Streamlit interface.

ACHIEVEMENTS

- **Winner, Pollution Forecasting track — AISEHack 2026** (IIIT Hyderabad, Apr 2026; ANRF-backed). Led team VibeCoders to forecast PM2.5 air pollution across India 16 hours ahead with a ConvLSTM model; proposed and added FADE (Fractional Advection–Diffusion Equation), which models how particles move in air.
- **SAS Curiosity Cup 2026 — Participant**. Built a Financial Stress Index in SAS from World Bank banking data using PCA; it predicted next-year banking stress with $R^2 = 0.879$.

CERTIFICATIONS

- **Deep Learning — IIT Ropar (NPTEL):** Elite + Silver, 87% • **Neo4j & Cypher Fundamentals** — Neo4j GraphAcademy